# Roshan Mathew Philip

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## Research Interests

### Mathematical Physics, Quantum Field Theories, Quantum Foundations

I am keenly interested in using mathematical structures to glean insights into physical theories and am particularly interested in working in mathematics that may lead to insights into quantum foundations, quantum gravity, or string theory. While I don't have any particular question in mind, I would like to work on a project that would let me experience mathematical physics research. Currently, I am reading up on Yang-Mills theory, and learning Conformal Field Theory.

## Experience

- Attended RADCOR 2025 - 17th International Symposium on Radiative Corrections: Applications of Quantum Field Theory to Phenomenology
  https://indico.niser.ac.in/event/2/ — Oct 2025
- Mathematical Foundations of Quantum Mechanics
  Guide: Anupam Pal Choudhury, School of Mathematical Sciences, NISER — Jan 2026 - Present
  Operator Theory, results for unbounded operators and infinite dimensional Hilbert spaces, spectral decomposition.
- Quantum Field Theory
  Guide: Shamik Banerjee, Schools of Physical Sciences, NISER — Feb 2025 - Present
  Scalar $\phi^4$ theory, QED, renormalisation, RG flow, Yang-Mills Theory, CFT
- Formalisms in Classical and Quantum Mechanics
  Guide: Sujay K. Ashok, Dept. of Theoretical Physics, IMSc — May 2024 - Jan 2025
  Canonical transformations in Hamiltonian mechanics; symplectic transformations; Poisson bracket formulation; path integral formalism in QM; resolvent, green's function and quantum scattering; homology groups, Lefschetz thimbles and Borel resummation.
- Non-Linear Dynamics and Chaos
  Guide: Sayantani Bhattacharya, School of Physical Sciences, NISER — Summer 2023
  Bifurcations, hysteresis and other phenomena in 2-D systems, rudimentary analysis of higher dimensional non-linear systems. Numerical analysis of the parameter space of the Lorenz Equations.
- Linear Dynamical Systems
  Guide: Anupam Pal Choudhury, School of Mathematical Sciences, NISER — Summer 2023
  An initial but rigorous foray into linear dynamical systems and bifurcations.

## Projects

- Defining a Metric on the Set of Images
- Percolation; Random Walks and Diffusion
- Exercises from Peskin and Schroeder QFT book
- Spherical model of a ferromagnet
- Magnon Bound States in 1d Heisenberg Chain
- Bead on a Inclined Rotating Hoop, Chaos and Resonances

## Achievements

- JEST-2026 exam **All India Rank 9**
- GATE-2025 exam **All India Rank 135**
- Innovation in Science Pursuit for Inspired Research (INSPIRE), Scholarship issued by Department of Science and Technology

## Skills

- **Relevant Topics that I have studied:**
  - Functional Analysis
  - Manifolds Theory and Differential Geometry
  - Measure Theory and Integration
  - Phase Transitions and Critical Phenomena
  - Quantum Field Theory and RG flows
  - General Theory of Relativity
  - Quantum Information and Computing
  - Quantum Optics
  - Programming (Embedded, and Numerical)
- **Currently learning:**
  - Non-Abelian Gauge Theory
  - Conformal Field Theory

## Talks and Sessions

- "*A Physicist's Introduction to Lie Groups and Algebras*" - NISER Physics Club, Nov 2025
- "*The N-Queens Problem*" - NISER Maths Club, 2022
- Informal tutoring for juniors in electromagnetism, nuclear physics, and particle physics.

## Extracurricular Activities

- Former Head of Coding Club, NISER (2024-26)
- Former Core Comittee Member of Physics Club, NISER (2023-24)
- Former Head of Chess Club, NISER (2024-25)
  FIDE rating of 1470.
- Piano
  Merit in Grade 5 Exam, Trinity School of Music, London.
- GM of a Dungeons & Dragons Campaign.

## Education

**National Institute of Science Education and Research,** Bhubaneswar, Odisha, India
Integrated BSc-MSc (A five year undergraduate cum master's course) 2022–Current
- Currently finished eighth semester
- CGPA: 9.23

**Chavara Public School** Kottayam, Kerala, India
CBSE Class XII 2022
- 96.4% with Physics, Mathematics, Computer Science, Chemistry

**St. Jude's Global School** Kottayam, Kerala, India
CBSE Class X 2020
- 96.6%